EDA MINI PROJECT REPORT

1. TOPIC: EXPLORATORY DATA ANALYSIS OF USED CARS

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DATASET SOURCE : USED CARS DATASET

2. OBJECTIVE

The objective of this project is to perform Exploratory Data Analysis (EDA) on a used-car dataset to understand its characteristics, identify missing values, and discover many other things.

The main objectives are:

- To understand the structures and features of the dataset.
- To identify missing values and duplicate values.
- To examine numerical and categorical variables.
- To analyse used-car prices, manufacturing years, fuel types, and transmission types.
- To identify outliers and understand data distributions.

MAIN QUESTIONS SELECTED FOR ANALYSIS

1.Which fuel type is most common in the dataset?

2.Which transmission type is more common?

3.Which locations contain the most car records?

4.How are used-car prices distributed?

5.Which manufacturing years are most represented?

3. DATASET OVERVIEW

The dataset used in this project contains information about used cars. It includes details such as car name, location, manufacturing year, fuel type, transmission type, kilometres driven, price and other features.

Dataset Summary

Feature	Details
Total rows	7,253
Total columns	14
Duplicate rows	0
Locations	11
Fuel categories	5
Transmission categories	2

Feature Descriptions

Column	Description	Data type
S.No.	Serial number	Integer
Name	Car name and model	Text
Location	Location of the car listing	Categorical
Year	Manufacturing year	Integer
Kilometers_Driven	Distance travelled	Numerical
Fuel_Type	Type of fuel	Categorical
Transmission	Manual or automatic	Categorical
Owner_Type	Ownership category	Categorical
Mileage	Car mileage	Numerical
Engine	Engine capacity	Numerical/text (before cleaning)
Power	Engine power	Numerical/text (before cleaning)
Seats	Number of seats	Numerical
New_Price	New-car price	Numerical/text (before cleaning)
Price	Used-car price	Numerical

Sample Records

S.No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price
1	Maruti Swift	Mumbai	2015	56,000	Petrol	Manual	First Owner	22.8	1197	86	5	6.7	4.5
2	Hyundai i20	Delhi	2016	41,000	Diesel	Manual	First Owner	24.0	1498	98	5	8.5	6.2
3	Honda City	Bangalore	2014	78,000	Petrol	Automatic	Second Owner	17.8	1497	118	5	9.8	7.5
4	Tata Nexon	Hyderabad	2018	32,000	Diesel	Manual	First Owner	23.5	1497	108	5	11.3	8.9
5	Ford EcoSport	Pune	2017	67,000	Petrol	Manual	Second Owner	16.5	1498	123	5	10.2	7.8

4.DATA QUALITY AND CLEANING

Data cleaning is necessary to improve the quality of the analysis.

Missing values

The dataset contains missing values in several columns.

S.No.	Column Name	Number of Missing Values	Percentage of Missing Values
1	New_Price	6247	86.15%
2	Price	1234	17.00%
3	Seats	53	0.73%
4	Engine	46	0.63%
5	Power	46	0.63%
6	Mileage	2	0.03%

The `New_Price` column contains the highest number of missing values. These missing values need to be considered when analysing prices.

Duplicate values

The dataset was checked for duplicate rows. The result was zero duplicate rows.

Outliers

Outliers were examined using box plots and the Interquartile Range (IQR) method. The manufacturing year was included in the outlier analysis.

Cleaning decisions

- Missing values were examined before analysis.
- Mode-based and median-based methods were used in the notebook for missing-value treatment.
- Duplicate records were checked.
- Outliers were examined.
- Numerical and categorical columns were considered separately during analysis.

5. DESCRIPTIVE ANALYSIS

Descriptive analysis helps summarise the important characteristics of the dataset.

Numerical Analysis:

It Summarizes the car data using mean, median, minimum, and maximum values.

Variable	Mean	Median	Minimum	Maximum
Year	2013.37	2014	1996	2019
Kilometers_Driven	58,699.06	53,416	171	6,500,000
Price	9.48	5.64	0.44	160

Categorical Analysis:

It Identifies the most common car location, fuel type, transmission, and owner type.

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Year	2013.37	2014	1996	2019
Kilometers_Driven	58,699.06	53,416	171	6,500,000
Price	9.48	5.64	0.44	160

6. Visual Exploratory Data Analysis (EDA)

Visual Exploratory Data Analysis (EDA) is used to understand the patterns and relationships in a dataset through charts and graphs. In this project, visual analysis is performed on the used cars dataset to understand car prices, manufacturing years, kilometers driven, fuel types, transmission types, and locations.

~ Car Price Distribution

A histogram can be used to show the distribution of used car prices. It helps us understand how many cars fall into different price ranges. The dataset contains both low-priced and high-priced cars. The mean price is approximately 9.48, while the median price is 5.64.

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~ Distribution of Cars by Manufacturing Year

A bar chart can represent the number of cars manufactured in different years. The manufacturing years range from 1996 to 2019, and the average manufacturing year is approximately 2013. This analysis helps identify the years that are well represented in the dataset.

~ Distribution of Kilometers Driven

A histogram can show how far the used cars have been driven. The median distance is 53,416 kilometers. The maximum recorded distance is 6,500,000 kilometers, which is unusually high compared with the median.

~ Distribution of Fuel Types

A bar chart can compare the number of cars using different fuel types. Diesel is the most common fuel type, with 3,852 records. This indicates that diesel cars make up a large portion of the dataset.

~ Distribution of Transmission Types

A bar chart can compare manual and automatic transmission cars. Manual transmission is the most common, with 5,204 records. This shows that manual cars are more frequently represented in this dataset.

~ Distribution of Cars by Location

A bar chart can display the number of used cars recorded in each location. Mumbai has the highest number of records, with 949 cars. This helps identify which locations are most represented in the dataset.

7. KEY FINDINGS AND LIMITATIONS:

1. Dataset Size: The dataset contains 7,253 rows and 14 columns, providing information about car models, locations, manufacturing years, fuel types, transmission, prices etc…

2. Manufacturing Year: The average manufacturing year is approximately 2013, and the median is 2014. The cars in the dataset were manufactured between 1996 and 2019.

3. Car Price: The `price` column contains 6,019 non-missing values. The mean price is 9.48, while the median is 5.64. This difference tells us that high-priced cars may increase the mean.

4. Kilometers Driven: The median distance driven is 53,416 km. The maximum value of 6,500,000 km is unusually high and requires further checking.

5. Fuel Type: Diesel is the most frequently recorded fuel type, with 3,852 cars.

6. Transmission: Manual transmission is the most common category, with 5,204 cars.

7. Ownership: First-owner cars form the largest ownership category, with 5,952 records.

8. Location: Mumbai has the highest number of records, with 949 cars.

9. Duplicate Records: No exact duplicate rows were found in the dataset.

8. CONCLUSION

In this mini project, Exploratory Data Analysis (EDA) was performed on a used cars dataset containing 7,253 rows and 14 columns. The main aim was to understand the dataset, examine its numerical and categorical features, identify missing values, and discover useful patterns through descriptive statistics and visualization

Overall, this project helped develop an understanding of the structure, quality, and main characteristics of the used cars dataset. The use of descriptive statistics and visualizations made it easier to summarize the data and identify patterns. The results can serve as a foundation for further analysis, such as studying how manufacturing year, kilometers driven, fuel type, transmission, and other car features relate to used-car prices.